

A computational baseline for changes in long-term evolvability

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Abstract

The concept of “evolvability” has long been debated in evolutionary biology; its definition, role, and evolution all lack consensus. In this work, we use one such definition: Evolvability is the long-term adaptive potential of a population. While this definition is fairly common, it creates issues of scale and metrics. How do you measure long-term potential or change in that potential? Here we build on previous evolution experiments that take evolvability at face value. We can run long-term evolution experiments to sample *what can evolve*, allowing us to estimate long-term potential. While such studies are prohibitively labor- and time-intensive for traditional wet-lab experiments, computational evolution can alleviate these barriers. We use hand-crafted landscapes to demonstrate that our method identifies the expected “evolvability-enhancing” mutations. We demonstrate that this definition of evolvability is dependent on factors outside the genotype-to-fitness mapping, such as mutation rate and population size. Finally, we employ this method to investigate how the number and fitness effect of evolvability-enhancing mutations change in more rugged fitness landscapes. We argue that computational evolution is a powerful tool in translating evolvability research from the conceptual to the empirical, and that we should further the mutualistic relationship between computational and wet-lab experiments of evolvability.

Introduction

How *well* will a population evolve in the future? This is the main question of *evolvability*, which concerns a population’s capacity for future evolution. Although evolutionary biologists have argued over the definition of evolvability for decades, there is still no agreed upon definition (Pigliucci, 2008; Nuño de la Rosa, 2017). Here we look beyond existing genetic variation and its effect on selection (Houle, 1992; Brown, 2014; Sterelny et al., 2007), and define evolvability as “the potential for adaptive evolution” (Wagner and Altenberg, 1996). Even after selecting a definition of evolvability, how do we measure it? While researchers have proposed multiple ways to measure evolvability (Houle and Pélabon, 2023; Hansen and Houle, 2008; Zaman et al., 2014) or differences in evolvability (Wagner, 2023), these methods are either theoretical, require many assumptions about the system, or are intractable to perform at scale. In this

work, we demonstrate computational evolution as a potential testbed for questions of evolvability. By leveraging the speed of evolution *in silico*, we directly measure long-term fitness potential of a population. With this information, we can experimentally test our assumptions of evolvability and verify other, more tractable, methods of measurement.

Here, we find changes in evolvability not by a complicated method, but by relying on our definition. We take heavy inspiration from Woods et al. (2011): If we view evolvability as long-term adaptive potential, why not directly measure that potential? The issue with this approach is scale. To approximate the adaptive potential of a population, we must evolve many replicates from the same initial condition. If we want to detect differences in evolvability, we must repeat this process for each condition we wish to compare. However, while evolving many replicates of *E. coli* might take months, evolving populations of digital organisms may take only seconds.

In this work, we position computational evolution as the missing step between theoretical understanding of evolvability and wet-lab experimental evolution (Draghi and Ogbunugafor, 2023). We first evolve populations on four hand-crafted, easily interpretable fitness landscapes (genotype-to-fitness mappings) to verify that our method correctly identifies the mutations designed to increase evolvability. To demonstrate that long-term fitness potential relies on factors outside the fitness landscape, we evolve replicates with different population sizes and mutation rates. We find that these changes can have drastic effects on evolvability, sometimes switching a mutation from significantly *increasing* evolvability to significantly *decreasing* evolvability. Finally, we demonstrate how these methods can be used to ask higher-level questions. We identify evolvability-enhancing mutations on increasingly rugged fitness landscapes, finding significant differences in the fraction of mutations that increase evolvability, but no difference in the fraction of evolvability-enhancing mutations that are beneficial. Overall, this work demonstrates that we can measure changes in evolvability by examining long-term fitness, and that these experiments are ripe for computational studies.

Background

While many definitions of evolvability exist (Pigliucci, 2008), here we leverage the definition of evolvability as “the potential for adaptive evolution” (Brown, 2014; Sterelny et al., 2007). This definition encompasses both standing genetic variation in a population upon which selection can operate (Houle, 1992) and the population’s capacity to generate such variation (Wagner and Altenberg, 1996). We also rely on this definition to sidestep the pitfalls of what constitutes a “developmental innovation” (Alberch, 1991) or “major evolutionary transition” (Smith and Szathmary, 1997). This definition only relies on the concept of a population’s fitness, a trait that applies in both computational systems and populations evolving in nature.

For this work, we are only concerned with a population’s *adaptive* potential. That is, we are looking for potential improvements in long-term fitness. We use adaptive potential as a starting point, as it is straightforward to measure in computational evolution systems. While some other researchers have limited their definitions of evolvability to adaptive changes (Payne and Wagner, 2019), evolvability can generally refer to any evolutionary potential (Brown, 2014; Hansen et al., 2019).

Much like the definition of evolvability, several metrics for quantifying evolvability have been proposed, but none are universally accepted. Building on early definitions of evolvability, researchers extended classical trait variance/covariance (G) matrices to include the effect of possible mutations (Jones et al., 2007) and selection on multiple traits (Hansen and Houle, 2008; Hansen et al., 2019). These methods, however, are typically restricted to short-term evolution and rely on the assumptions of the underlying Lande equation (Houle and Pélabon, 2023). Looking at relative differences instead of raw evolvability, Wagner (2023) sought to identify “evolvability-enhancing” (EE) mutations by isolating mutations whose one-step mutational neighborhood has higher fitness than the wildtype. Again, this analysis only detects short-term evolvability. Finally, Brown (2014) mathematically defined evolvability as the probability of evolving a focal feature given an initial population and environment state. This method, however, suffers from *how* to calculate that probability.

Experimental evolution provides one additional avenue of inquiry: empirical studies of evolution in the lab or in controlled field studies (Kawecki et al., 2012). Currently, few studies have directly tested theories of evolvability using experimental evolution. However, many studies have empirically investigated factors likely to contribute to evolvability, while others discuss evolvability as it becomes relevant during an experiment (for reviews, see Payne and Wagner (2019); Barnett et al. (2025); Draghi and Ogbunugafor (2023)).

Within experimental evolution, one method is particularly relevant to discussions of evolvability. Leveraging the

frozen “fossil record” of *E. coli* from the Lenski Long-term Evolution Experiment (LTEE) (Lenski et al., 1991), Blount et al. (2008) used analytic replay experiments to experimentally identify mutations that increased the lineage’s *potential* to evolve citrate metabolism. Expanding on this idea, Woods et al. (2011) again replayed LTEE lineages, but compared their long-term fitness instead of the potentiation of a given trait, directly measuring the lineages’ adaptive potential. Indeed, other researchers have conducted similar experiments to quantify evolvability (McBride et al., 2008; Barrick et al., 2010). These experiments range from 50 to 883 generations and evolved replicates stemming from at most 8 initial genotypes/strains. Although these studies give us unprecedented glimpses into long-term evolvability, they come at the cost of substantial time and labor.

Computational evolution provides an alternative to wet-lab experiments that are too time- or labor-intensive. Indeed, one of the first uses of the term “evolvability” stems from artificial embryology (Dawkins, 1988). Computational evolution studies have previously identified conditions that lead to short-term evolvability (e.g., a higher percentage of mutations that cause adaptive phenotypic change) (Zaman et al., 2014; Canino-Koning et al., 2016; Crombach and Hogeweg, 2008). Other studies have investigated the evolvability of development patterns (Draghi and Wagner, 2008) and selection of evolvability (Earl et al., 2004) using computational models. Recent work in artificial life has demonstrated that computational evolution allows for studies of evolutionary potential at scales intractable in natural organisms (Ferguson and Ofria, 2023; Renner et al., 2024).

Methods

In this work, we demonstrate how computational evolution can precisely identify genetic changes that increase evolvability. Here we use NK landscapes, simple fitness landscapes that allow us to control the degree of “ruggedness”. We perform three experiments: 1) we verify our method by identifying evolvability-enhancing mutations in handcrafted fitness landscapes; 2) we demonstrate that evolvability, as we define it, is dependent on factors outside of the fitness landscape (population size and mutation rate); and 3) we identify evolvability-enhancing mutations in randomly generated NK landscapes to determine how landscape ruggedness affects evolvability.

Detecting changes in evolvability

Using our definition of evolvability (the potential for adaptive evolution), we take inspiration from Woods et al. (2011) – we experimentally estimate a population’s potential by running multiple evolutionary replicates from the same starting condition (here 100 replicates with different random seeds) and observing the distribution of evolved fitness. By varying the starting conditions (e.g., different initial populations or different mutation rates), we can determine the

effect that change has on the evolved fitness distribution.

As is often the case with evolvability, our evolved fitness distributions depend on context, and they are not universally comparable. Using this method, we cannot directly compare the evolvability of *E. coli* and our bitstring organisms. We can, however, detect changes in evolvability within a system. We can consider two treatments, as described by the starting conditions of evolution replicates. One treatment is our unaltered baseline, and the other treatment alters one variable: population genetic makeup, mutation rate, or population size. After conducting a substantial number of evolutionary replicates for each treatment, we are left with a distribution of final maximum fitness for each treatment. Since we have no guarantee of normality, we detect significant increases in fitness distribution (and thus evolvability) using a one-sided Mann-Whitney-Wilcoxon test (Mann and Whitney, 1947). We correct for multiple comparisons using the Bonferroni method.

Evolution system

Even in computational evolution, it can take hours or days to evolve populations in complex environments. For this work, we use NK landscapes due to their explainability and speed. NK landscapes provide a “tunably rugged” genotype-to-fitness mapping for bitstring organisms (Kauffman and Weinberger, 1989). N denotes the length of the bitstrings. When considering a bit’s contribution to fitness, you also include the next K bits. This creates epistasis between the bits and thus increasing K increases the ruggedness of the fitness landscape.

All experiments used roulette (fitness proportional) selection to select organisms for the next generation (Goldberg and Deb, 1991). Generations were synchronous and populations were well-mixed (i.e., no spatial structure). Unless otherwise stated, populations contained 1,000 organisms and organisms had a mutation rate of 0.1% per bit. All replicates evolved for 1,000 generations.

Experiment 1: Validation in hand-crafted landscapes

We first verify that our method detects mutations that increase evolvability under scenarios where evolvability dynamics are obvious. In each landscape, we run evolutionary replicates with clonal populations of each possible genotype. This allows us to enumerate *all* evolvability-enhancing mutations.

We analyze four hand-crafted landscapes:

1. **Hillclimbing:** One global optimum, which is reachable via adaptive climbs from any genotypes. Evolvability should be constant across the landscape. A mutation toward the optimum is always a 10% increase in fitness.
2. **Equal gradients** We construct two fitness gradients of equal steepness (10% fitness gain per step) that are con-

nected at the bottom. One gradient has six mutations while the other has three, creating a global and local optimum, respectively. Mutations that increase the probability that the population reaches the global optimum should increase evolvability.

3. **Neutral region** We keep the same local and global optima from Landscape 2. Instead of fitness gradients, optima are separated by a region of neutral mutations. Although they are neutral, mutations that bring the population closer to the global optimum should increase evolvability.
4. **Valley crossing** A local optimum is separated from the global optimum by a deleterious fitness valley. Mutations that increase the probability of crossing the valley should also increase evolvability. Each step into the valley is a 5% reduction in fitness, but the global optimum has a ~46% increase in fitness over the local optimum, equivalent to that many steps along the gradient from Landscapes 1 and 2.

All hand-crafted landscapes have a K equal to $N-1$, which allows for easy construction of exact fitness values (Ferguson and Sanson, 2025). For each landscape, we constructed both a one-dimensional and a full N -dimensional NK landscape. One dimensional landscapes are considerably easier to visualize and parse, but we included the full landscapes as assurance that this method works in larger, more complicated landscapes. The one-dimensional case was constructed by setting most genotypes in the landscape to lethal (zero fitness). However, this lowers the viable mutation rate of populations evolving on these one-dimensional fitness landscapes. After considering the differences in mutation rate (which is directly tested in Experiment 2), evolvability results did not fundamentally change between the one-dimensional and N -dimensional landscapes. Due to space, we only show the one-dimensional results here, but all results are available in the supplement (Ferguson and Sanson, 2025).

Experiment 2: Effects of non-landscape factors

Evolvability is typically considered an attribute of a *population*. Experiment 1 varies one aspect of a population: its genetic makeup. When we define evolvability as a population’s adaptive potential, we must consider what other factors contribute to that potential. Here, we vary two properties of the population, its size and the underlying mutation rate, to see how they affect evolvability. Population size is known to alter the effects of selection and genetic drift (Wright, 1931), while mutation rate is often discussed as affecting evolvability (Lynch et al., 2016; Bedau and Packard, 2003). Thus, any baseline of evolvability should respond to changes in either population size or mutation rate.

To test these factors, we repeat Experiment 1. We evolve populations on the same landscapes but vary the population

sizes and mutation rates. For each landscape, we evolve 100 replicates per genotype with a per-bit mutation rate of 0.01%, 0.05%, 0.1% (the same from experiment 1), 0.5%, and 1%. Similarly, with our base mutation rate of 0.1%, we evolve 100 replicates with 10, 100, 1,000 (same as experiment 1) and 10,000 organisms. After evolving these populations, we will compare which mutations increase evolvability under each treatment. We then qualitatively analyze how the set of evolvability-enhancing mutations changes as we vary these two population attributes, to see how factors beyond the fitness landscape affect evolvability.

Experiment 3: Characterization of fitness effects

Last, to demonstrate a use case of these techniques, we analyze the effect of landscape ruggedness on the number of evolvability-enhancing mutations. We also analyze the fitness effects (i.e., beneficial, neutral, or deleterious) of evolvability-enhancing mutations.

We revert to full (but small) randomly-generated NK landscapes (Kauffman and Weinberger, 1989), where the number of bits in a genotype, N , is six. We vary the ruggedness of the landscape, K , and detect evolvability-enhancing mutations using the method described above. We again evolve 100 replicates starting from each possible genotype, with all replicates evolving for 1,000 generations. Once identified, we record 1) the fraction of possible mutations that increase evolvability and 2) the fraction of evolvability-enhancing mutations that are beneficial. We then conduct a Kruskal-Wallis test to determine if significant variation exists across all treatments (Kruskal and Wallis, 1952).

Data and software availability

Our model was built using the Modular Agent Based Evolver 2 (MABE2) software (<https://github.com/mercere99/MABE2>). Code, analyses, and data are available in the supplemental material (Ferguson and Sanson, 2025). Analyses, statistics, and plots used R version 4.1.2 (R Core Team, 2021) with the dplyr (Wickham et al., 2022), ggplot2 (Wickham et al., 2020), cowplot (Wilke, 2020) and ggbeeswarm (Clarke et al., 2023) packages.

Results and discussion

Evolvability depends on fitness landscape topography

As expected, all populations of standard size and mutation rate evolve to reach the global optimum in the hillclimbing landscape, regardless of their initial composition (Figure 1, red dashed line). Because all populations evolve to the same fitness, no mutations increase evolvability on this simple landscape.

In Landscape 2 (equal gradients) we see differences in long-term fitness potential (Figure 2, red dashed line). 55 of the 100 replicates seeded from the genotype at the base of

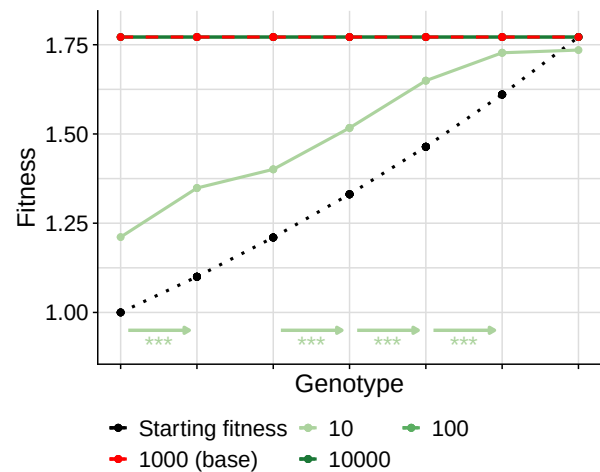


Figure 1: **Landscape 1: Single gradient.** Lines show the fitness of the seven possible genotypes (dotted black line) and average evolved fitness from 100 populations starting at that genotype. The red dashed lines show base conditions (Experiment 1), while solid lines show other population sizes (Experiment 2). All populations greater than size 10 evolved to the global fitness maximum. See Figure 2 for a full description.

the gradients evolved to the global optimum. If the population was seeded even just one step onto a fitness gradient, all replicates evolved to optimum at the end of that gradient. Thus we see two mutations that significantly ($p < 0.001$) enhance evolvability: the mutation to move from the local optimum's gradient to the base of the gradients, and the mutation onto the global optimum's gradient. This matches our expectation that mutations that increase the probability of evolving along the longer gradient increase evolvability.

In the neutral landscape (Landscape 3, red dashed line), populations founded with genotypes within two mutations from an optimum always evolved to reach that optimum (Figure 3). Populations founded from genotypes three mutations away from the global optimum evolved to it in 99 of 100 replicates, which was not a significant decrease from two mutations away. In contrast, populations started three mutations away from the local optimum only reached it in 93 replicates, which means that the mutation from three to two steps from that optimum was weakly significant ($p < 0.05$). Indeed, populations founded in the middle of the neutral region often failed to reach either optimum in the allotted 1,000 generations (i.e., they stayed in the neutral region), and thus all mutations that moved them toward an optimum increased evolvability. These results generally matched our expectation. This landscape demonstrates that 1) mutations toward a local optimum can increase evolvability if that local optimum is better than other alternatives and 2) neutral

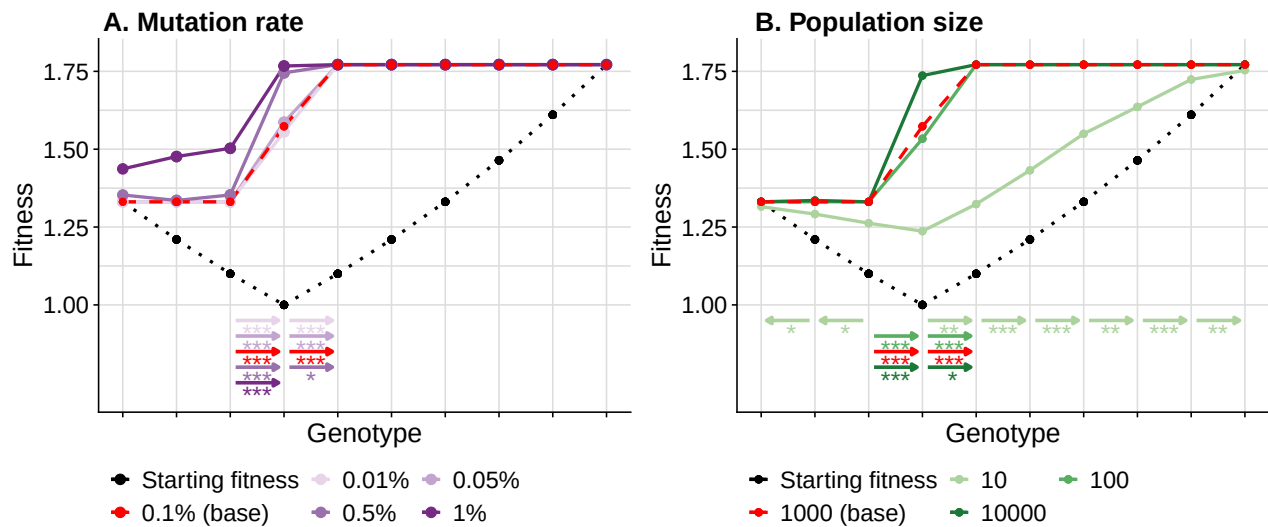


Figure 2: **Landscape 2: Equal gradients.** The x-axis shows the ten possible genotypes in the one-dimensional equal gradients landscape. Adjacent genotypes are mutational neighbors. The dotted black line shows the fitness of each genotype, while the other lines show average evolved fitness of 100 populations starting at that genotype. The dashed red line is shared between panels A and B, showing the evolved fitness under base conditions (Experiment 1). The solid lines show evolved fitness under different per-bit mutation rates (panel A) and population sizes (panel B) from Experiment 2. Arrows at the bottom show mutations that significantly increase evolvability. Arrows are ordered in increasing mutation rate / population size. Significance in the Mann-Whitney test (with Bonferonni corrections) is shown as * ($p < 0.05$), ** ($p < 0.01$), and *** ($p < 0.001$).

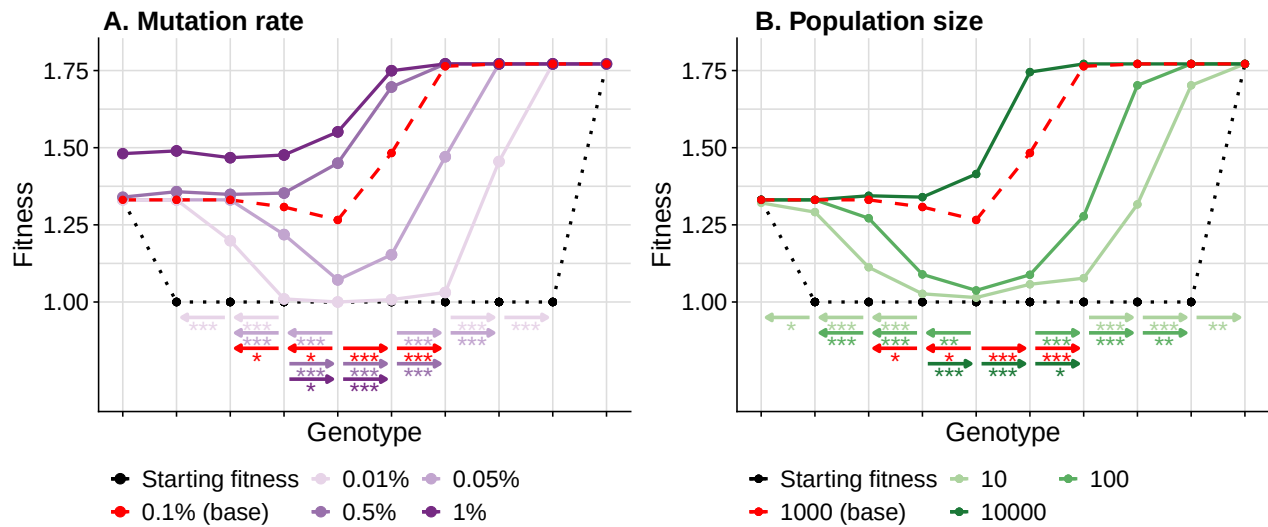


Figure 3: **Landscape 3: Neutral space.** See Figure 2 for a full description.

mutations can increase evolvability.

Finally, populations founded before or one step into the fitness valley of Landscape 4 never crossed the valley (Figure 4). Populations founded two steps into the valley crossed in 43 of 100 replicates, and populations founded three steps into the valley always crossed. Thus, mutations that moved the population deeper into the fitness valley were highly significant ($p < 0.001$). This result aligns with previous studies

that demonstrate the importance of deleterious mutations in long-term evolution Covert III et al. (2013) and how later steps into a fitness valley drastically increase the probability of crossing that valley (Ferguson et al., 2024).

These results demonstrate that we can detect changes in evolvability even in simple fitness landscapes, and that our method detects changes that we intuitively expected to increase evolvability. We have shown that beneficial, neu-

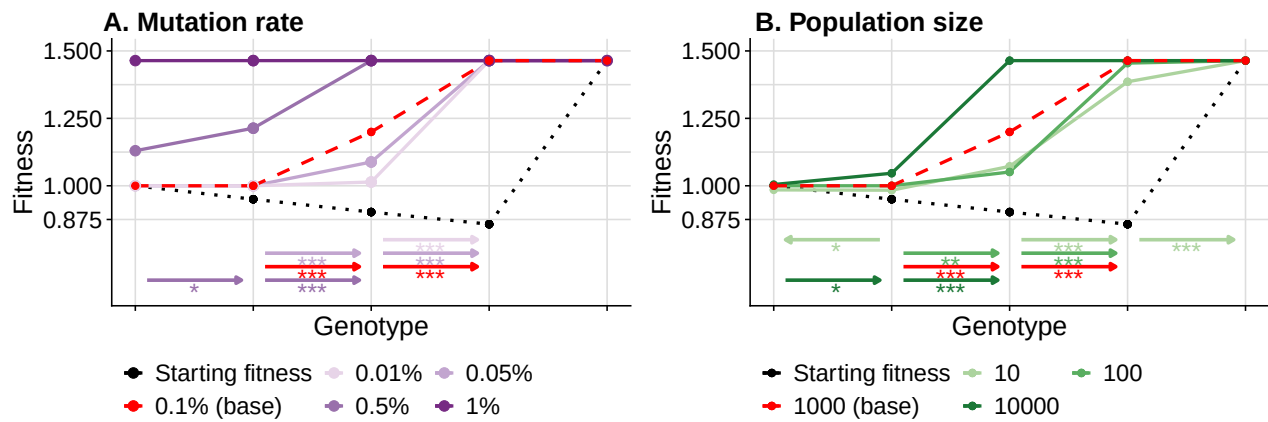


Figure 4: **Landscape 4: Valley crossing.** See Figure 2 for a full description.

tral, and deleterious mutations can all increase evolvability, which matches related studies (Woods et al., 2011; Covert III et al., 2013).

This analysis also demonstrates how short-term analyses, such as those that look at one-step mutational neighborhoods (Wagner, 2023; Canino-Koning et al., 2016), can identify different mutations as increasing evolvability. The equal gradients landscape (Landscape 2), demonstrates that a mutation may increase short-term fitness while also limiting long-term fitness by increasing the probability of fixation at a local optimum.

Mutation rate and population size can drastically alter evolvability

In the hillclimbing landscape, the mutation rates we tested (0.01% through 1% per bit), had no effect on evolvability. Indeed, all but one population evolved to the global optimum regardless of the mutation rate. Similarly, all populations of 100 or more organisms evolved to the global optimum (Figure 1). Populations of size 10, however, often failed to evolve to the global optimum. Some populations that seeded at the global optimum even drifted to sub-optimal genotypes. In this small population, four of the six possible beneficial mutations increased evolvability. While a population of size 10 is unrealistic in most natural populations, this demonstrates that population size can influence the evolvability dynamics of the population.

Populations evolving on the equal gradients landscape demonstrate that both mutation rate and population size can change which mutations increase evolvability. As mutation rate increases, populations from across the landscape become more likely to reach the global optimum. While the lower mutation rates see two evolvability increasing mutations, the highest mutation rate (1% per-bit) has only one. 99 of 100 populations started from the genotype between the two gradients evolved to the global maximum, so the first mutation onto the longer gradient did not meaningfully

change the evolvability of the population. A similar trend is seen with population size, as populations founded from that central genotype are very likely to evolve to the global optimum. However, even at our maximum population size, the first step onto the longer gradient still increases evolvability ($p < 0.05$). Finally, the weakened selection pressure and increased genetic drift of the size 10 population means that mutations toward either optima increase evolvability.

Increases in mutation rate and population size had similar effects in the neutral landscape. For low mutation rates and small populations, mutations that bring the population close to an optimum significantly increase evolvability. As mutation rate or population size increase, it is the central mutations – those that gently bias the population to one end of the neutral region – that increase evolvability. One mutation *completely flips direction*, increasing evolvability with a mutation to the left at low rates / small sizes, but increasing evolvability with a mutation to the right in other treatments. For the largest population size and highest mutation rates, populations are guaranteed to find *one* of the optima. Thus only mutations that bias the population toward the global optimum's end of the neutral region significantly increase evolvability.

A population can only cross a deleterious fitness valley if a clade that mutates into the valley can accumulate enough mutations before purifying selection removes it. Therefore we expect changes that increase mutational inflow, such as larger population sizes and higher mutation rates, to generally increase the probability that the population crosses the valley. This is exactly what we see; as we increase population size or mutation rate, populations are able to cross the valley from earlier steps into the valley. We observe that evolvability-enhancing mutations occur earlier into the valley as either of these values increases. For the highest mutation rate tested, there are no evolvability-enhancing mutations because *all* populations cross the fitness valley. Again, the purifying selection is so low in the populations of size

10 that a mutation *backward* out of the valley is also a significant increase in long-term fitness ($p < 0.05$).

These results suggest that our method is capable of detecting how factors outside the fitness landscape affect evolvability. Here we are qualitatively comparing which mutations significantly increase long-term fitness potential (evolvability) across mutation rates or population size. However, the method should also handle the opposite experiment: locking which genotype we are analyzing and comparing long-term evolved fitness when we vary population size or mutation rate. This is crucial, as it would allow the analysis of how “mutator” genotypes, which increase effective mutation rate, affect evolvability. These mutator genotypes are theorized to play a key role in long-term evolution, with support from both wet-lab populations (Sniegowski et al., 1997) and computational studies (Taddei et al., 1997; Bedau and Packard, 2003).

Landscape ruggedness creates more evolvability-enhancing mutations

As a case study of how these methods can be used to ask theoretical questions of evolvability, we generated random NK landscapes ($N=6$) of increasing ruggedness (K). We then used our method to identify evolvability-enhancing (EE) mutations to quantify A) what fraction of all possible mutations increase evolvability and B) what fraction of EE mutations were beneficial.

Unsurprisingly, we found that random landscapes with no ruggedness ($K = 0$) had no mutations that increased evolvability (Figure 5A). These landscapes are randomized hill climbs, so all populations should reach the optimum regardless of founding genotype. This result is consistent with Landscape 1 above.

As ruggedness increases, we see that only two landscapes with $K=1$ have evolvability-enhancing mutations. In those two landscapes, 14.2% and 22.4% of mutations significantly increased evolvability. By $K=2$, all but one landscape contains EE mutations, and all landscapes with ruggedness $K \geq 3$ have EE mutations. When we average the fraction of mutations that increase evolvability within each ruggedness value, we see this mean increase to a peak of 33.4% at $K=4$. We detect significant differences in the fraction of mutations that increase evolvability across values of K ($p < 0.001$, Kruskal-Wallis).

We then isolated the evolvability-enhancing mutations and determined if they were beneficial or deleterious. We did not differentiate between beneficial/deleterious and weakly beneficial/weakly deleterious, and no mutations were truly neutral. We found that the fraction of EE mutations that were beneficial varied by landscape in [22.8%, 59.2%]. This excludes one outlier at $K=3$ where a landscape had only five EE mutations, all of which were deleterious. A Kruskal-Wallis test failed to detect significant differences between K values ($p > 0.4$).

This experiment provides evidence that many, if not most, mutations that increase evolvability are deleterious. We agree with (Wagner, 2023) that these deleterious EE mutations are less likely to be selected than beneficial EE mutations. However, if we restrict discussions of evolvability to only neutral and beneficial mutations, we risk overlooking critical mutations that can drastically increase long-term fitness (Covert III et al., 2013). While deleterious mutations are unlikely to fix in a population, they may produce a further-mutated offspring before being eliminated by purifying selection. That double-mutant offspring could be beneficial compared to the original wildtype (i.e., sign epistasis).

Conclusion

In this work we used computational evolution as a bridge between conceptual discussions of evolvability and experimental evolution. We leveraged a “long-term adaptive potential” definition of evolvability and employed techniques similar to Woods et al. (2011) to empirically sample those long-term fitness distributions. Using the speed of computational evolution, we enumerated evolvability across one- and N -dimensional fitness landscapes, empirically creating “evolvability landscapes”. We were then able to identify *all* mutations that significantly increased evolvability.

Using this method for identifying evolvability-enhancing (EE) mutations, we conducted two experiments to verify the method and a one to serve as a usage case study. First, we constructed easily-intuited fitness landscapes and demonstrated that the method is agnostic to fitness effects and can identify EE mutations that are neutral, deleterious, or beneficial. Second, we examined two factors of an evolving population that exist outside the fitness landscape: population size and mutation rate. We verified that, even in these simple models, we can identify how these two factors alter evolvability. Indeed, these methods could provide valuable insight into how mutations such as “mutator alleles” alter evolvability (Taddei et al., 1997), and computational systems offer speed and experimental control that are perfect for initial experiments. Finally, our third experiment was a case study that investigated how the ruggedness of the fitness landscapes affects EE mutations. We found that increasing ruggedness increases the number of EE mutations, but found no evidence that the proportion of EE mutations that are beneficial changes. While these simple computational systems lack the complexity of biological organisms, we can use them to ask evolvability questions that are currently intractable in wet labs.

The landscapes analyzed in this work were intentionally simple. Decades of articles and arguments clearly indicate that evolvability is a complex topic that will require much more work to understand. Our goal in this work is to position computational evolution as an early stepping stone in that path. The speed of computational evolution allows for rapid prototyping and experiments of evolvability; instead

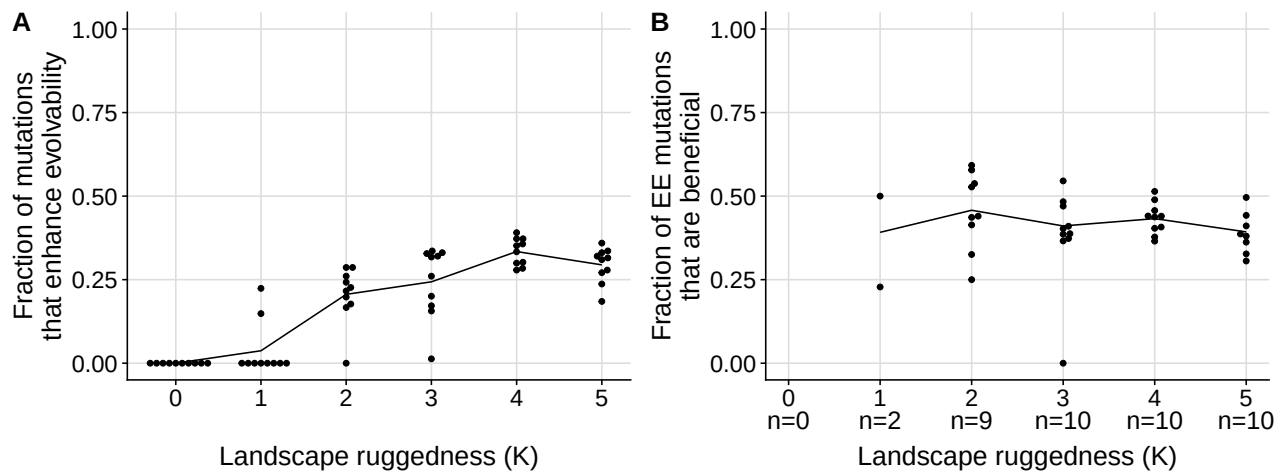


Figure 5: **Experiment 3.** Panels show details of evolvability-enhancing (EE) mutations in NK landscapes of increasing ruggedness. Ten random NK landscapes were analyzed for each value of K. Panel A shows the fraction of EE mutations out of all possible one-step mutations. The line shows the overall fraction for that value of K, while the fraction of EE mutations in each of the 10 landscapes is shown via beeswarm plot. Panel B shows the fraction of those EE mutations that are beneficial. The number of landscapes that have EE mutations, n, is shown below the x-axis.

of requiring months of laboratory effort, a computing cluster allows us to enumerate the evolvability of all genotypes in these simple landscapes in less than an hour. We see the utility of these computational studies as twofold: 1) to create baselines of how evolvability changes in increasingly-complex fitness landscapes, and 2) to refine our methods and metrics of measuring evolvability to save time and effort in wet-lab experiments. Our understanding of evolvability cannot come from computational experiments alone, but they can provide the conceptual and methodological groundwork upon which more realistic experiments are built.

Limitations and future work

There are many limitations to this study; some were intentional to keep the experiments simple while others are open questions in the field. The landscapes in this study were not only simple, but incredibly small (a maximum of 64 genotypes). In such small, bounded landscapes, evolution is likely to reach an optimum in very few generations. Indeed, our hand-crafted landscapes were built around comparing *which* optimum a population evolves to. However, evidence suggests that the fitness landscapes of natural organisms are not only larger, but *open-ended*, even in controlled laboratory experiments (Wiser et al., 2013). Open-ended evolution has been a goal of artificial life researchers for decades, but no systems have reached a consensus of truly open-ended evolution (Taylor et al., 2016). Even if they are not truly open-ended, similar experiments should be conducted in more complex systems (e.g., Avida, Aevol) to test these methods in systems closer to natural organisms.

This work *enumerated* long-term fitness for all genotypes

in the landscape. In a more complex system or larger landscape, this becomes intractable. However, these methods work for comparing the evolvability of *any* combination of genotypes (Woods et al., 2011). Enumerations studies like this work will provide valuable insights into *which* genotypes should be compared in more complex systems.

Wagner (2023), focuses on beneficial mutations when searching for evolvability-enhancing mutations, as deleterious EE mutations are unlikely to be selected. In this work we demonstrated that deleterious mutations can significantly increase evolvability. Our methods, however, do not account for the *probability* of a mutation being selected, as we always seed full populations of a single genotype. Future work should investigate different ways of seeding these populations. If all genotypes come from a lineage, population snapshots offer a nuanced view of long-term evolvability over time (Ferguson et al., 2024). In other scenarios, populations could be founded primarily from a wildtype genotype with one or a few organisms containing the focal mutations. These technique would provide a more realistic view of how a mutation, occurring in small portion of the population, might affect long-term evolution.

The ultimate goal of this work is to help bridge the gap between conceptual understandings and experimental studies of evolvability (Draghi and Ogbunugafor, 2023). Future work should refine these techniques, use them as a baseline for proxy measurements of evolvability, and ultimately deploy them in wet-lab studies to better our understanding of life around us.

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References

- Alberch, P. (1991). From genes to phenotype: Dynamical systems and evolvability. *Genetica*, 84(1):5–11.
- Barnett, M., Meister, L., and Rainey, P. B. (2025). Experimental evolution of evolvability. *Science*, 387(6736):eadr2756.
- Barrick, J. E., Kauth, M. R., Streltsov, C. C., and Lenski, R. E. (2010). *Escherichia coli* rpoB Mutants Have Increased Evolvability in Proportion to Their Fitness Defects. *Molecular Biology and Evolution*, 27(6):1338–1347.
- Bedau, M. A. and Packard, N. H. (2003). Evolution of evolvability via adaptation of mutation rates. *Biosystems*, 69(2):143–162.
- Blount, Z. D., Borland, C. Z., and Lenski, R. E. (2008). Historical contingency and the evolution of a key innovation in an experimental population of *Escherichia coli*. *Proceedings of the National Academy of Sciences*, 105(23):7899–7906.
- Brown, R. L. (2014). What Evolvability Really Is. *The British Journal for the Philosophy of Science*, 65(3):549–572.
- Canino-Koning, R., Wiser, M. J., and Ofria, C. (2016). The Evolution of Evolvability: Changing Environments Promote Rapid Adaptation in Digital Organisms. In *Proceedings of the Artificial Life Conference 2016*, pages 268–275, Cancun, Mexico. MIT Press.
- Clarke, E., Sherrill-Mix, S., and Dawson, C. (2023). Package ‘gg-beeswarm’. *Categorical scatter (violin point) plots*.
- Covert III, A. W., Lenski, R. E., Wilke, C. O., and Ofria, C. (2013). Experiments on the role of deleterious mutations as stepping stones in adaptive evolution. *Proceedings of the National Academy of Sciences*, 110(34):E3171–E3178.
- Crombach, A. and Hogeweg, P. (2008). Evolution of Evolvability in Gene Regulatory Networks. *PLOS Computational Biology*, 4(7):e1000112.
- Dawkins, R. (1988). The evolution of evolvability. In *Artificial Life*, pages 201–220. Routledge.
- Draghi, J. and Wagner, G. P. (2008). Evolution of Evolvability in a Developmental Model. *Evolution*, 62(2):301–315.
- Draghi, J. A. and Ogbunugafor, C. B. (2023). Exploring the expanse between theoretical questions and experimental approaches in the modern study of evolvability. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*, 340(1):8–17.
- Earl, D. J., Deem, M. W., and Chandler, D. (2004). Evolvability Is a Selectable Trait. *Proceedings of the National Academy of Sciences*, 101(32):11531–11536.
- Ferguson, A. J. and Ofria, C. (2023). Potentiating Mutations Facilitate the Evolution of Associative Learning in Digital Organisms. In *ALIFE 2023: Ghost in the Machine: Proceedings of the 2023 Artificial Life Conference*. MIT Press.
- Ferguson, A. J., Ofria, C., and Bohm, C. (2024). Predicting the Unpredictable: Using replay experiments to disentangle how evolutionary outcomes are altered by adaptive momentum. In *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*. MIT Press.
- Ferguson, A. J. and Sanson, M. (2025). Supplemental material (github). URL: <https://doi.org/10.5281/zenodo.16891740>.
- Goldberg, D. E. and Deb, K. (1991). A Comparative Analysis of Selection Schemes Used in Genetic Algorithms. In Rawlins, G. J. E., editor, *Foundations of Genetic Algorithms*, volume 1, pages 69–93. Elsevier.
- Hansen, T. F. and Houle, D. (2008). Measuring and comparing evolvability and constraint in multivariate characters. *Journal of Evolutionary Biology*, 21(5):1201–1219.
- Hansen, T. F., Solvin, T. M., and Pavlicev, M. (2019). Predicting evolutionary potential: A numerical test of evolvability measures. *Evolution*, 73(4):689–703.
- Houle, D. (1992). Comparing evolvability and variability of quantitative traits. *Genetics*, 130(1):195–204.
- Houle, D. and Pélabon, C. (2023). Measuring evolvability. In *Evolvability: A Unifying Concept in Evolutionary Biology?*, pages 101–119. MIT Press.
- Jones, A. G., Arnold, S. J., and Bürger, R. (2007). The Mutation Matrix and the Evolution of Evolvability. *Evolution*, 61(4):727–745.
- Kauffman, S. A. and Weinberger, E. D. (1989). The NK model of rugged fitness landscapes and its application to maturation of the immune response. *Journal of Theoretical Biology*, 141(2):211–245.
- Kawecki, T. J., Lenski, R. E., Ebert, D., Hollis, B., Olivieri, I., and Whitlock, M. C. (2012). Experimental evolution. *Trends in Ecology & Evolution*, 27(10):547–560.
- Kruskal, W. H. and Wallis, W. A. (1952). Use of Ranks in One-Criterion Variance Analysis. *Journal of the American Statistical Association*, 47(260):583–621.
- Lenski, R. E., Rose, M. R., Simpson, S. C., and Tadler, S. C. (1991). Long-term experimental evolution in *Escherichia coli*. I. Adaptation and divergence during 2,000 generations. *The American Naturalist*, 138(6):1315–1341.
- Lynch, M., Ackerman, M. S., Gout, J.-F., Long, H., Sung, W., Thomas, W. K., and Foster, P. L. (2016). Genetic drift, selection and the evolution of the mutation rate. *Nature Reviews Genetics*, 17(11):704–714.
- Mann, H. B. and Whitney, D. R. (1947). On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other. *The Annals of Mathematical Statistics*, 18(1):50–60.
- McBride, R. C., Ogbunugafor, C. B., and Turner, P. E. (2008). Robustness promotes evolvability of thermotolerance in an RNA virus. *BMC Evolutionary Biology*, 8(1):231.

- Nuño de la Rosa, L. (2017). Computing the Extended Synthesis: Mapping the Dynamics and Conceptual Structure of the Evolvability Research Front. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*, 328(5):395–411.
- Payne, J. L. and Wagner, A. (2019). The causes of evolvability and their evolution. *Nature Reviews Genetics*, 20(1):24–38.
- Pigliucci, M. (2008). Is evolvability evolvable? *Nature Reviews Genetics*, 9(1):75–82.
- R Core Team (2021). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Renner, A., Marois, É., Fiorito, J., Ashworth, J., and Yoder, J. A. (2024). A Computational Model of Developmental Exaptations. In *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*. MIT Press.
- Smith, J. M. and Szathmari, E. (1997). *The Major Transitions in Evolution*. OUP Oxford.
- Sniegowski, P. D., Gerrish, P. J., and Lenski, R. E. (1997). Evolution of high mutation rates in experimental populations of *E. coli*. *Nature*, 387(6634):703–705.
- Sterelny, K. et al. (2007). What is evolvability. *Philosophy of biology*, pages 163–178.
- Taddei, F., Radman, M., Maynard-Smith, J., Toupance, B., Gouyon, P. H., and Godelle, B. (1997). Role of mutator alleles in adaptive evolution. *Nature*, 387(6634):700–702.
- Taylor, T., Bedau, M., Channon, A., Ackley, D., Banzhaf, W., Beslon, G., Dolson, E., Froese, T., Hickinbotham, S., Ikegami, T., McMullin, B., Packard, N., Rasmussen, S., Virgo, N., Agmon, E., Clark, E., McGregor, S., Ofria, C., Roppella, G., Spector, L., Stanley, K. O., Stanton, A., Timperley, C., Vostinar, A., and Wiser, M. (2016). Open-Ended Evolution: Perspectives from the OEE Workshop in York. *Artificial Life*, 22(3):408–423.
- Wagner, A. (2023). Evolvability-enhancing mutations in the fitness landscapes of an RNA and a protein. *Nature Communications*, 14(1):3624.
- Wagner, G. P. and Altenberg, L. (1996). Complex Adaptations and the Evolution of Evolvability. *Evolution*, 50(3):967–976.
- Wickham, H., Chang, W., Henry, L., Pedersen, T. L., Takahashi, K., Wilke, C., Woo, K., Yutani, H., and Dunnington, D. (2020). *Ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics*.
- Wickham, H., François, R., Henry, L., and Müller, K. (2022). *Dplyr: A Grammar of Data Manipulation*.
- Wilke, C. O. (2020). *Cowplot: Streamlined Plot Theme and Plot Annotations for Ggplot2*.
- Wiser, M. J., Ribeck, N., and Lenski, R. E. (2013). Long-Term Dynamics of Adaptation in Asexual Populations. *Science*, 342(6164):1364–1367.
- Woods, R. J., Barrick, J. E., Cooper, T. F., Shrestha, U., Kauth, M. R., and Lenski, R. E. (2011). Second-order selection for evolvability in a large *Escherichia coli* population. *Science*, 331(6023):1433–1436.
- Wright, S. (1931). Evolution in Mendelian Populations. *Genetics*, 16(2):97–159.
- Zaman, L., Meyer, J. R., Devangam, S., Bryson, D. M., Lenski, R. E., and Ofria, C. (2014). Coevolution drives the emergence of complex traits and promotes evolvability. *PLoS Biology*, 12(12):e1002023.